جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

ENGLISH LANGUAGE DEPARTMENT (ELD)

ENGL101

READING BOOK PORTFOLIO: PART 2

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Introduction to Active Reading

Important notes and benchmark dates to keep in mind:

Actively read texts 6-10 by the end of week 8. One text in the Reading Final exam will be taken from these texts.

*You will shortly be provided a comprehensive guide on what active reading entails and how you are supposed to practically implement active reading strategies. Please be sure to refer back to these guidelines throughout the semester in order to give yourself the best opportunity to achieve higher marks for your Reading Portfolio grade.

*Writing while you read and taking notes enables you to become an active reader and is an essential part of understanding a text.

What is Active Reading?

Active reading refers to the mental processes that highly effective readers use when approaching reading. It requires a reader to read critically by focusing on the material to understand and actively engage with the material by being aware of one's own thought process when reading. Through active reading readers gain greater critical thinking skills that makes things easier to understand and enables readers to retain information for a longer period.

Active Reading: General Strategies

Annotating

Marking and highlighting a text during reading is a way for readers to stay engaged with a reading and comprehend the reading at a deeper level. Readers record responses to the text during the act of reading and write ideas and personal comments in the margins.

Previewing

Readers look over the reading material to become familiar with topic and organization before actually beginning to read it. This helps make the reading an easier, faster, and more effective learning experience.

Skimming

The reader looks quickly through material to gain an overall view of text without reading everything.

What is Active Reading sourced and adapted from MPC reading skill sheets https://www.mpc.edu/academics/academic-divisions/humanities-division/programs-centers/reading-center/handouts.

Scanning

The readers look through a text to locate specific information without reading everything, such as names, dates, and figures/tables.

Reading on

Readers skip an unfamiliar word and read further to provide sufficient context needed to determine the meaning of an unknown word.

Re-Reading

A good reader re-reads the text again for deeper understanding, word identification and fluency.

Monitoring Comprehension

Readers recognize when they don't understand parts of a text and take necessary steps to restore meaning. Monitoring includes asking clarifying questions if something remains unclear, re-reading if there is some confusion, looking for answers and adjusting reading strategy to understand material.

Determining Importance

Readers distinguish between what information in a text is most important versus what information is interesting but not necessary for understanding. Knowing the primary purpose of reading textbooks and nonfiction is important to determine importance.

Author's Purpose

Being aware of the author's purpose allows a reader to understand the reason and intent of the writing.

Paraphrasing

Readers re-state and re-write text in their own words to capture the author's main idea and details of the reading. This strategy forces readers to pay close attention to the author's ideas and helps improve the readers' level of understanding.

Summarizing

Readers reduces larger texts to focus on important elements by identifying key elements and condensing important information into their own words to solidify meaning.

Active Reading Strategies sourced and abridged from MPC reading skill sheets https://www.mpc.edu/academics/academic-divisions/humanities-division/programs-centers/reading-center/handouts.

Questioning

Readers engage with the text by asking questions about the text and the author's intentions. They then seek information to clarify and extend their thinking before, during and after reading.

Synthesizing

Readers piece together and combine information from their own knowledge of how life works with the information they have gathered and understood from the text to create their own perspective and original insight.

Evaluating

The reader forms an opinion and judgment about the writing. They determine whether the argument (if there is one) is well or poorly structured, supported, or detailed; whether the writing style is appropriate, efficient, or well-toned--whether the audience is appropriately addressed; or if the author is biased in any way and how this may have affected the writing.

Active Reading: Examining a Text

When studying a text closely, consider the following points. Note that reading exam questions often test your ability to understand these elements of a text:

1. What is the text about / the main idea?

Look at the title, sub-title and any pictures, graphs or tables

Read the introduction. Look for a thesis statement (a sentence showing what the article is about). Note that in some academic texts this could be in the second paragraph.

Skim the paragraphs. What are most paragraphs about? The balance of information often reflects the main purpose of the article.

Read the conclusion. It should reaffirm your thoughts on the main idea of the article.

Note that in test conditions, some students prefer to answer these types of questions last after reading the text in more detail from answering the other questions.

2. What type of text is it?

Is the writer just reporting the facts of news, an event or research; or are they describing the causes and / or effects of something; or are they providing a comparison or giving an opinion?

Where would you see this type of text? The topic, length and use of vocabulary should indicate if it is an article aimed at the average reader (e.g. a newspaper article) or the more academic reader (e.g. a science journal).

What tense is used? Does it change? Why?

3. What is the main idea of each paragraph?

It's important, especially in test situations, to understand the main idea of each paragraph. Look for the topic sentence (usually first or second).

The main idea should be supported with an explanation and / or examples.

4. Does the writer include facts (statistics, dates, people)?

If so, what do these statistics mean? It's important to understand them. If graphs or tables are provided with a text, it's vital to read and understand them.

If dates are given, why? They often indicate something important.

If people are mentioned, why? Are they experts, witnesses or giving opinion? Again, it's important to consider why individuals or organizations are mentioned in a text and what role they play.

5. Does the writer suggest or imply anything?

Sometimes a writer suggests something (e.g. an opinion) without saying it directly. Can you see any examples in the text?

6. Is there any unknown vocabulary?

Again, underline any unknown/unsure vocabulary. Read the sentences before and after the word. Does any information help to guess the meaning of the word?

Key:

Main idea
Useful information
Writing
Vocab

TEXT 5

Global Climate Change: Can wind and solar power play a role?

Blew, T. (2012, July) Global climate change: Can wind and solar power play a role? *World of Renewables*. Retrieved from www.greenpowerworld.com

The first paragraph introduces the topic for the reader. The essay will talk about if the solar and wind power can be useful.

Ever increasing worldwide use of oil, gas, and coal—fossil fuel—leads to more carbon dioxide (CO_2) emissions, which in turn causes global warming. In addition, fossil fuel is a finite resource. This means that in the long run the world needs to find alternatives to oil, gas, and coal—the fossil fuels. The power sources that replace fossil fuels must be cleaner, and not add to pollution or climate change. Some environmentalists argue that wind and solar power are the answer. However, when evaluating the claims of the environmentalists, three factors have to be taken into consideration: ______X______, ______Y______, ______Z______.

limited

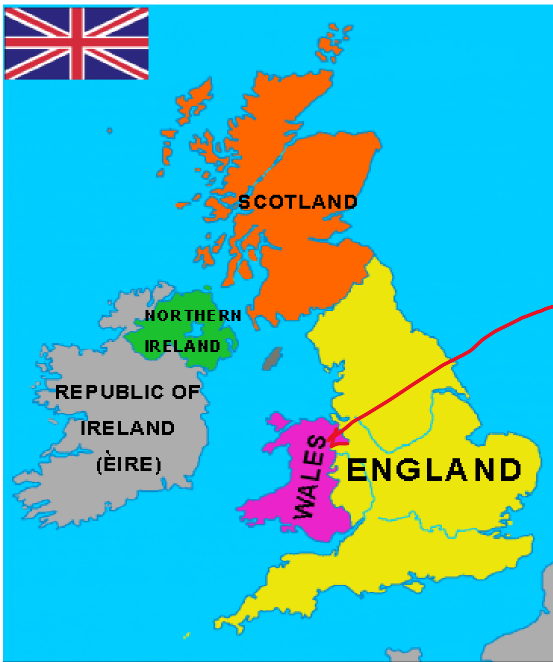


Figure 1. Map of the United Kingdom (UK)

That represent a huge area of land and it will just provide 15% of the needed energy which means that we will need more than six times of Wales to get the energy needs for UK.

Power density

In terms of power density, wind and solar power compare very badly with other renewables and fossil fuels. Power density, with respect to energy sources, refers to the energy flow that can be harnessed from a given area and is measured in watts per meter squared (W/m^2). While wind and solar have a power density of 1.2 and 6.7 W/m^2 respectively, fossil fuels and nuclear power have a power density of at least 100 W/m^2, and some power plants considerably higher.

Explanation of power density

Easy to be done

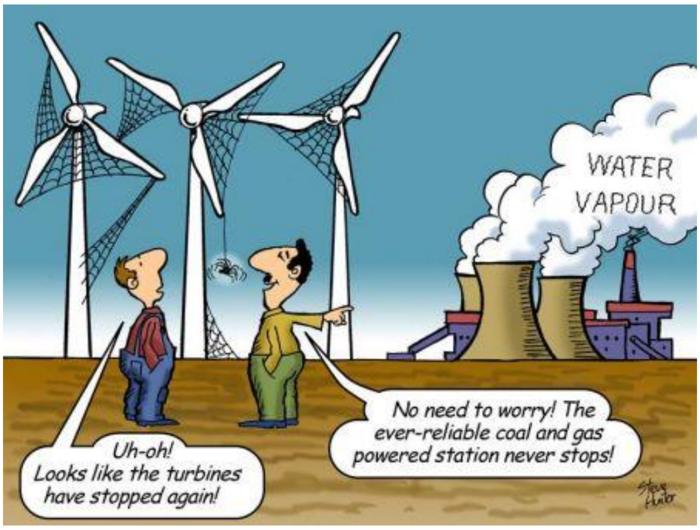
What does this mean with regard to the practicalities and feasibility of using wind or solar power as one of our energy sources? It means that for wind or solar power to make an appreciable contribution to meeting our energy needs, huge areas of land would have to be set aside to locate the requisite number of turbines or solar panels. For example, to generate around 15% of the UK's energy needs an area the size of Wales (see Fig 1.) would be needed to house the 100,000 turbines required. With regard to area, and cost, this is unfeasible. Another example pertains to biofuels. Biofuels have a power density of around 0.5-2 W/m2. To meet the world's current energy consumption of 16 TW (16 trillion watts), we would have to convert around 50% of the land surface (75 million km2) to biofuel production.

Large enough to be noticed

necessary

Related to

Not possible



Intermittency

Unexepted change

Another disadvantage of wind and solar power is that they are dependent on the *vagaries* of the weather. If the wind does not blow or the sun does not shine, there is no energy produced. On one occasion in the summer, figures show that the output of three UK wind farms was *sufficient* only to power three electric kettles. Another feature that must be *factored in* is that wind turbines and solar panels rarely operate at 100% efficiency. On many days the wind velocity is *insufficient* for the wind turbine to meet its optimum speed. Also, rather perversely, on days with high velocity winds, the turbines have to be 'switched off' to prevent mechanical damage. As regards solar panels, cloud cover and a build-up of dirt on the panels can considerably reduce efficiency.

Enough

consider

Not enough

Support

Because of the reasons outlined above, if we have to depend on one of these intermittent sources of energy, we must have a *backup*, or reserve. If these were other renewable energy forms, intermittency of wind and solar would be a relatively *minor* criticism. However, much of this backup energy is provided by fossil fuels in the form of coal, oil or natural gas fired power stations. These cannot be started up and closed down at short notice, so they have to be 'running' continuously at a low level in the event they are needed. Therefore, even when we are relying on wind and solar power, we are still *emitting* quantities of carbon dioxide. Unfortunately, these fossil fuels are perfect partners for these *erratic* renewable energies.

Produce

unpredictable

Storage

Another fundamental problem with wind and solar power is storage. With fossil fuels the fuel carries the energy 'within it' in the easily *accessible* form of chemical energy. We burn the fuel and release the energy, as and when we need it. Wind and solar, in contrast, act independently of demand. Energy is produced when the wind blows or the sun shines. This energy is available immediately. If it is not used, it is lost. This inability to store the electricity

Able to reach

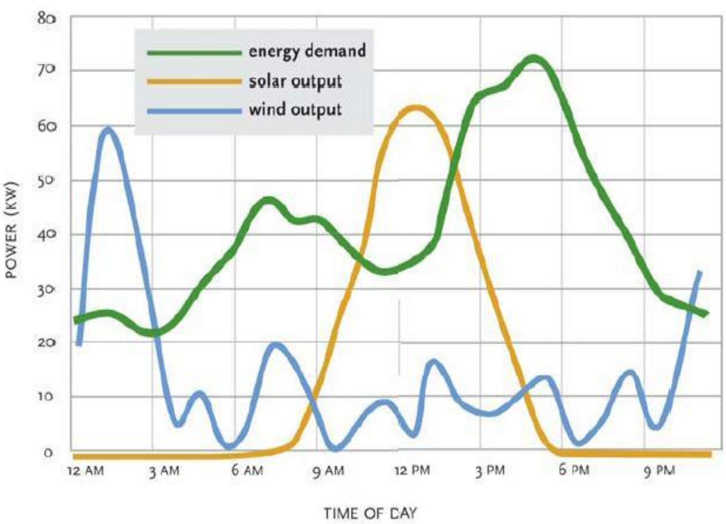


Figure 2. What does Figure 2 tell us?

The figure shows us that the wind and solar energy are not constant, they are produced more in some times and less in the other.

produced by wind power is one of the biggest factors holding back the drive for more wind and solar power.

The same

An illustration of the problem of storage is the case of Denmark. For a small country Denmark has some of the largest wind farms. Unfortunately, the periods when wind conditions are ideal usually coincides with the periods when the demand is at its lowest. Consequently, Denmark has to sell this excess energy to its neighbors at a much reduced rate. When the demand in Denmark rises, it has to buy energy back from its neighbors at peak rates. As a result, Danish consumers are hit with some of the world's highest electricity charges.

More than necessary

Author Tim Blew spent 25 years as an engineer specializing in renewable and non-renewable energy technologies. In the last 7 years he has worked predominantly on Carbon Capture and Storage (CCS) technologies and Enhanced Oil Recovery (EOR) techniques. He also acts as an advisor to the Carbon Alliance, an organization funded by companies in the fracking industry. The remit of the Alliance is 'to provide objective, scientific evidence to facilitate a balanced debate about our energy future'. He has written for a number of well-respected magazines such as The New Scientist, The Economist and Scientific American. He is also a regular contributor to many news organizations, including the BBC and CNN.

If you were writing a report about Saudi Arabia's energy future, would you accept the author Tim Blew as a source of information? Write a short paragraph explaining your decision.

Yes, because he has the knowledge about renewable and non-renewable energy so he is reliable. Also, he has the ability to convence the people as in the text he provide a very clear reason about why we should not rely on solar and wind energy.

1. Look at the introductory paragraph, what does X, Y and Z refer to?

It refers to the thesises and it tell the reader how many body baragraph the essay have and what is the topic of each one. X: Power density Y: Intermittency Z: Storage

True, False, Not stated	**Highlight the correct answer**
Paragraph A	
2. ~~Four~~ 3 types of fossil fuel are mentioned in Paragraph (A).	True, **False**, Not stated
3. Carbon Dioxide (CO_2) is the main greenhouse gas.	True, False, **Not stated**
4. CO_2 emissions from fossil fuels are the main cause of climate change.	**True,** False, Not stated
Paragraph B	
5. Power Density (PD) is the energy produced in a given area of land.	**True**, False, Not stated
6. Power density is measured in W/m3.	True, **False,** Not stated
7. Solar power has a higher PD than wind.	True, False, Not stated
8. Biofuels are a type of fossil fuel.	**True**, False, Not stated
9. Biofuels are more efficient than solar power.	True, **False**, Not stated
Paragraph C	
10. Solar power is more reliable than wind power.	True, **False**, Not stated
11. The UK has stopped using wind power because it is so inefficient.	True, False, **Not stated**
12. High winds often result in no wind energy production.	**True,** False, Not stated
13. Solar power efficiency can be reduced due to dirt on the panels.	**True**, False, Not stated
Paragraph D	
14. Even when relying on wind and solar, CO_2 is still produced.	**True,** False, Not stated
15. Fossil fuels store energy as mechanical energy.	True, **False**, Not stated
16. An inability to store their energy is the biggest problem of wind and solar.	**True**, False, Not stated
17. Due to wind power, Denmark has the lowest electricity prices in Europe.	True, False, **Not stated**

Part B: Sentence completion. Based on the text, complete the sentences.

1. Because fossil fuels are a finite resource, we cannot rely on it to obtain our energy because it will end one day.
2. Due to the problems of climate change and pollution, some environmentalists claim that using solar and wind power can prevent that.
3. In comparison to nuclear power and fossil fuels solar and wind energy cannot provide an enough energy and they need a loyt of areas for tha panels and torbins.
4. Power density can be explained as the energy flow that can be used from a given area.
5. Huge areas of land would be needed due to the low power density solar and wind have.
6. Due to the poor PD of wind it requiers a huge area of land for the turbines and that makes it unefficinte.
7. An area the size of Wales will be needed to provide 15% of the energy requires in UK.
8. Approximately 50% of the Earth's surface should be transform into biofuel production to provide the needed energy for the current world.
9. Wind turbines rarely operate at maximum speed because they can cause abig damege if they work with high wind.
10. A back up energy source is often needed when using solar or wind power and it is always fossil fuel.
11. While we can burn fossil fuels as and when we need energy , wind and solar power need sun and wind to performe well and give us the energy we need.
12. Denmark has to buy energy from its neighbors when the wind is not enough to generate the adequte eneregy.

Go back through the text, find the words in *blue italics* and find a synonym or short explanation for what they mean.

Look at Figure 2. Write a short paragraph explaining the graph and its implications.

The graph illustarte how much solar and wind power are prodiced in every time in the day. It is obvious that the wind power is prodiced more from 12am to 3am and the solar power is at maximum at arond 12pm. While in the middle of the day there are few energy and that cannot match the demand in the energy. Moreover there are some times in the graph where there is no enough soalr and wind power and the demand is high. We can conclude from the graph that we cannot depend on solar and wind power as they cannot provide the energy demand.

Using what you have learned from the text, and your own personal knowledge, write a proper paragraph about the feasibility of renewable energies (wind and solar) for KSA.

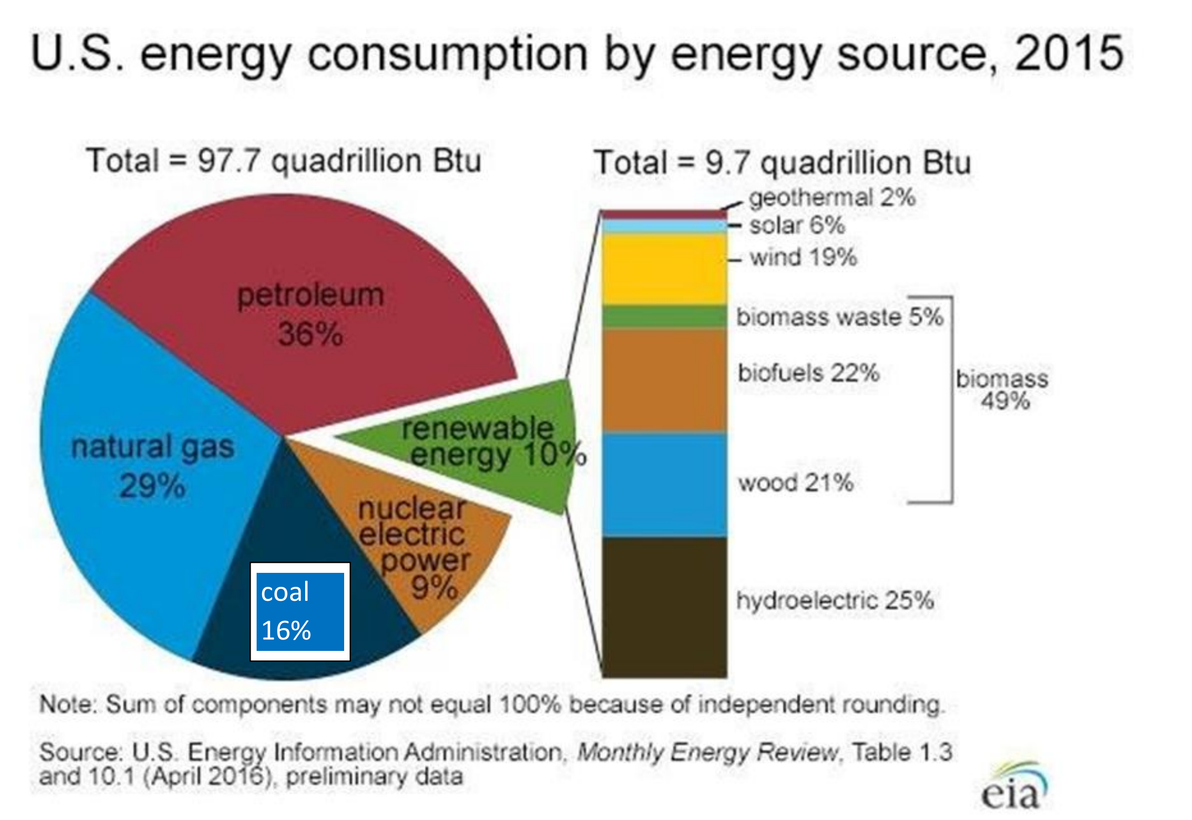
Alghtho Saudi have the biggest gas and petrol fieald, Saudi Arabia are moving to reneable energy and that is becuase petrol and gas are limited resorces. Using solar power would be so efficnt in the summer as the tempressure is high and night is short. On the other hand, Wind power may not be smart to use as Saudi does not have a lot of wind and ussally when there is a wid it is dirty and that can affect the turpins. If Saudi use the solar power and reduce the use of gas and petrol that will generate cleaner energy and that can develop the cuntry.

TEXT 6: Pre reading skills:

Why nuclear is the only way forward.

Let's have a look at the visuals and see what they tell us about the text.

Figure 1.



1. What percentage of US energy consumption come from fossil fuels?

   81%

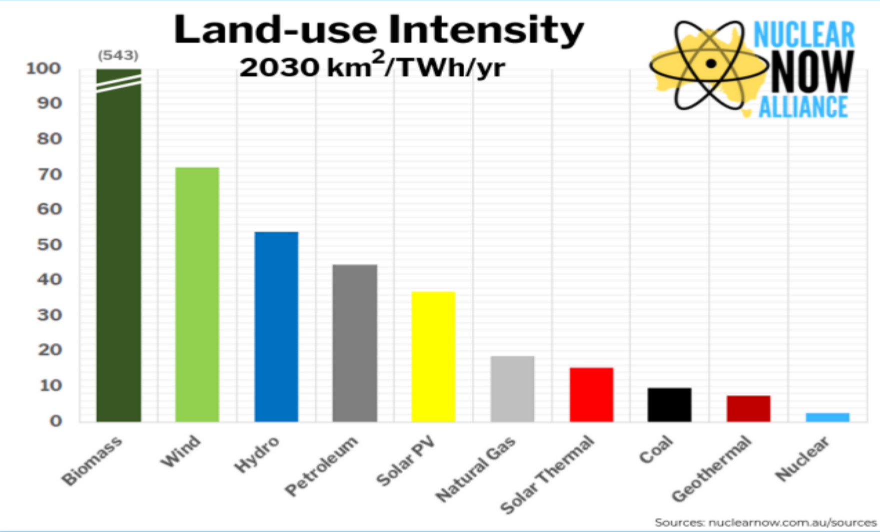
2. What percentage of US energy consumption come from renewables?

   10%

3. What is the author's purpose in including this diagram?

   To show how much US use every energy source and it is clear that most of the energy is from fossil fules.

Figure 2. Land-use intensity



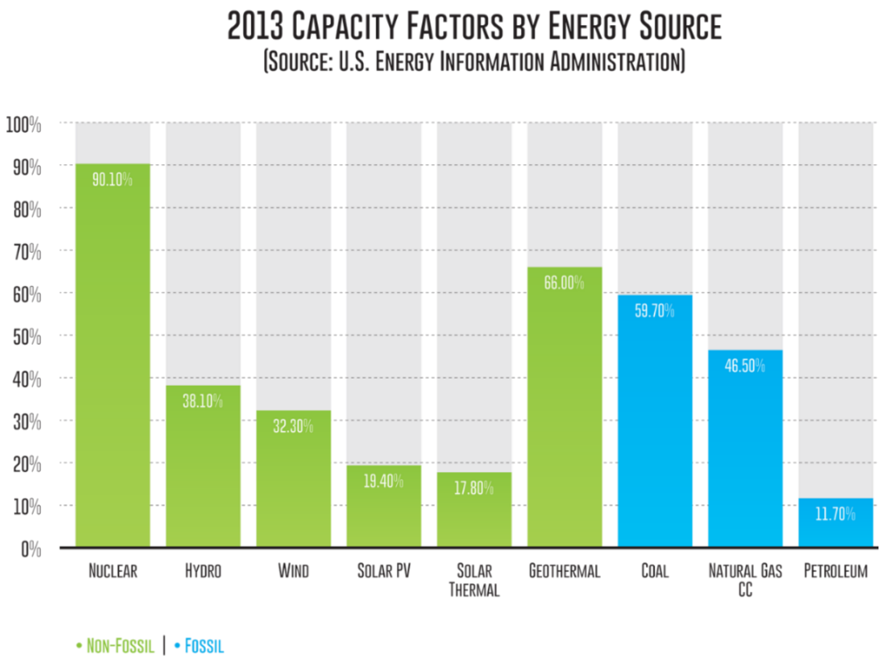
What is capacity factor? You probably do not know, but let's see what we can take from the chart. Which energy sources have the highest and lowest capacity factors? Is it better to have a high or low capacity factor? Make an 'educated guess', and we can check our 'guess 'later.

the nuclear has the highest capacity factor and the petroleum has the lowest.

I think it is better to have high capacity factor because it will help to reduce the land needed.

4. Which energy sources require the most/least amount of land?

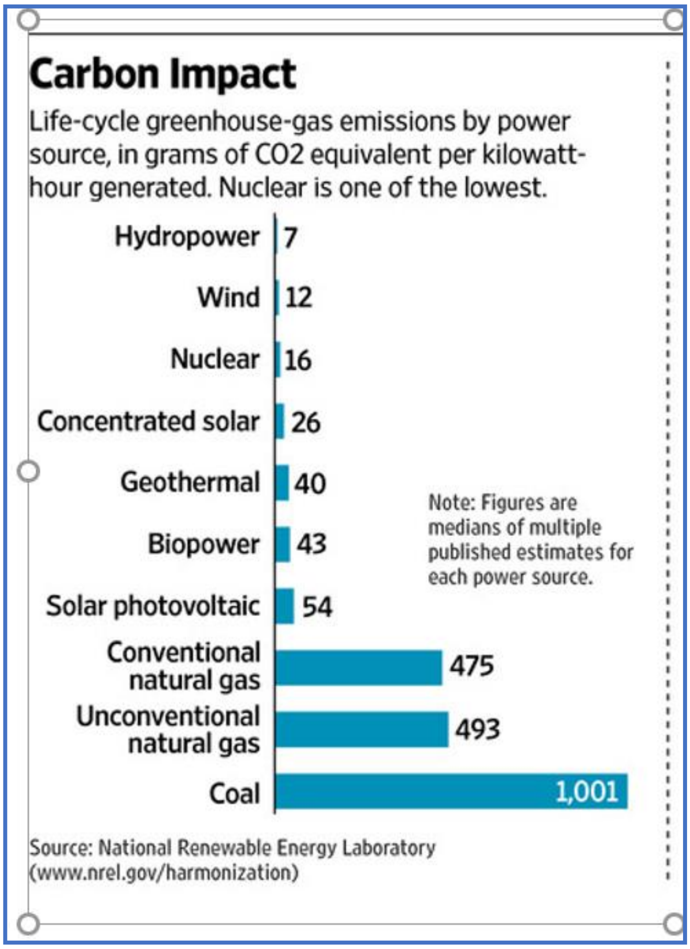
Biomass require the most and nuclear require the least.



Briefly summarize what Figure 4 illustrates.

The figure shows the Carbon impact on diffrent energy sources. The hydropwer is the lowest and the coal is the highest.

Figure 4



If you have been paying attention, you should be able to answer this question. Why does power density matter?

Because we want an energy that can give the most with the smallest use of land.

5. What is the author's purpose in including this figure?

Figure 3 shows the capacity factor and the writer include it to say that the nuclear has the largest capacity and that make it efficint.

6. Where did the author get the information from? Any thoughts or comments on this?

From the national renewable energy laboratory. It shows that the author is relaible as he take his informaition from relaiable websites.

Power Density and why it matters

Identify the Article subheadings below

Capacity factor

Any ideas what this means? We may have to wait until we read the text.

It is the actual electricity production over maximum possible electricity output in a time.

CO2 emissions

What will be discussed in this section?

The different type of energy and how much CO2 it provides.

Nuclear Waste

In this paragraph, the author will tell us how dangerous nuclear waste is. Do you agree or disagree? Explain your answer.

Yes, I agree that the waste of nuclear is a negative effect of using it, but there is a solation for that even if it is expensive it will allaw us to get the benifits of nuclear energy.

Based on the title, sub-titles, and visuals, what do you think the text will be about? Write a brief paragraph.

The text will probably talk about nuclear energy and why we sholde think about it as the main source in the future. The graphs show us that the nuclear energy have a lot of strengthen and it does not provide a lot of CO_2 emmision.

Now read the introduction and the topic sentences. How accurate was your prediction? What did you get wrong? Why do you think you got it wrong?

My gusses were true, as the world is seaching for an energy sorce that can provide energy to this populated world without emmions, the nuclear whould be the best choice for the future.

Key:

Main idea
Useful information
Writing
Vocab

Why nuclear is the only way forward.

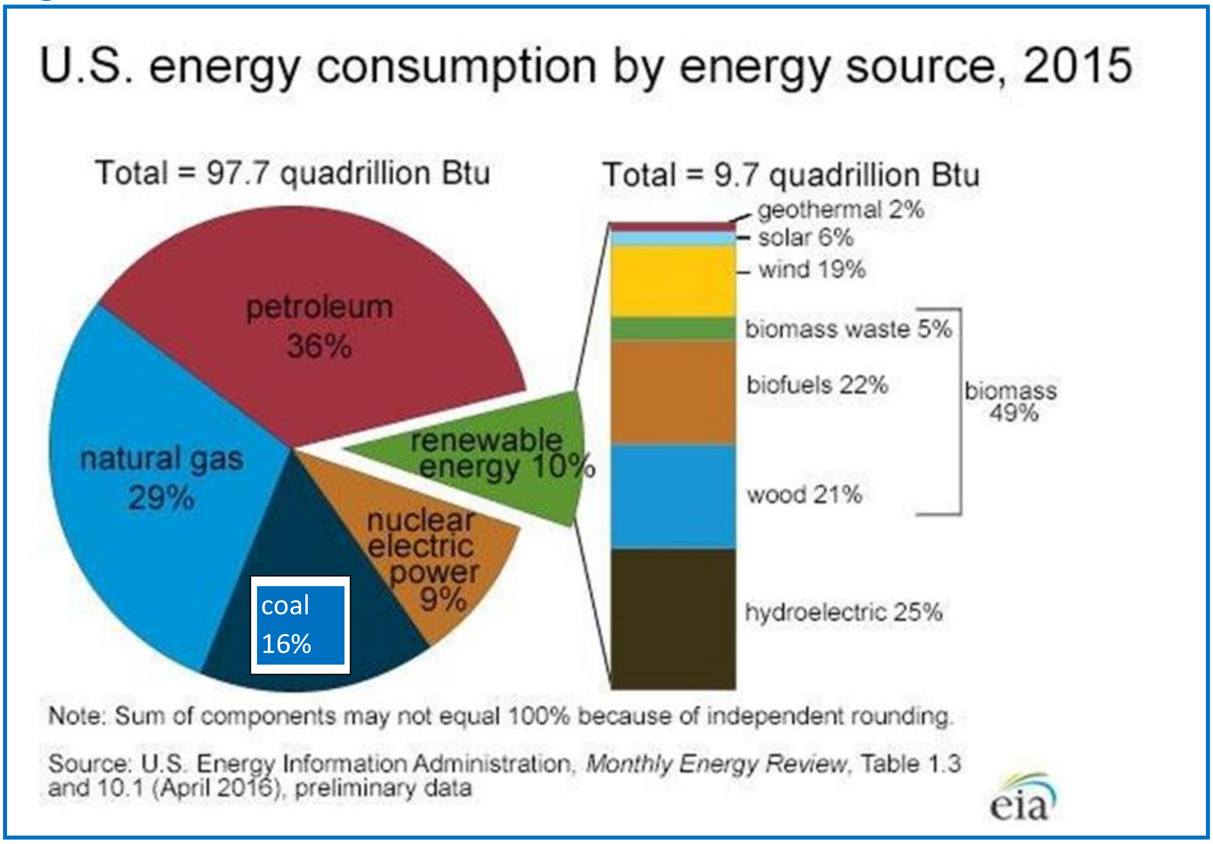
Richards, T. (2020, May 12). Why nuclear is the only way forward. *The Daily Record*. Retrieved from <http://www.dailyrecord.co.uk/debate/article-2713830/Why-Nuclear-is-the Way-Forward.html>. (composite source for classroom use)

(A) Climate change caused by the emissions of greenhouse gases from the burning of fossil fuels is now widely accepted by the scientific community as the most serious of the environmental problems the world is facing. It is this fact that provides the **impetus** for our efforts to *transition* away from fossil fuel use to 'greener' sources of energy. The list of renewable energy sources offered up as alternatives is a long one. However, despite the massive investment in renewables in the form of research and development and subsidies to encourage use, renewables still only make up 10% of energy consumption in the USA (World energy consumption is a similar figure). If renewables only meet 10% of our energy needs now, what chance have they got once we factor in population increase and the resulting rise in energy demand, which is predicted to increase by 50% by the year 2050. This 'inconvenient truth' illustrates that if we wish to move away from energy produced by fossil fuels, the only source that can produce vast quantities of energy without greenhouse gas emissions is nuclear power.

Change from one state into another.

The author starts with talking about climate change to say that we have to search for an energy that does not increase the climate change but reduce it.

Figure 1.



Power density and why it matters

(B) With respect to renewable energy, there are limitations to the amount of energy we can extract from a given area of land. Most forms of renewable energy have very low power densities. (See Table 1 for comparisons).

Table 1. Power densities of some common energy sources

* Figures are averages from multiple estimates

largest

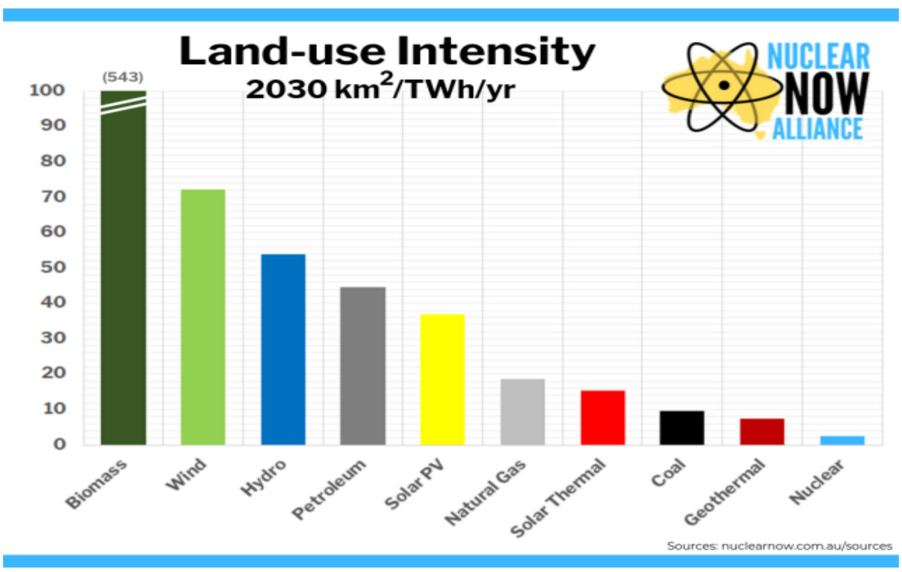
Wind	Solar	Coal	Oil	Nuclear
2.5 - 11 W/m2	6.7 – 15 W/m2	500 - 1000 W/m2	1000 W/m2	2000 - 6000 W/m2

lowest

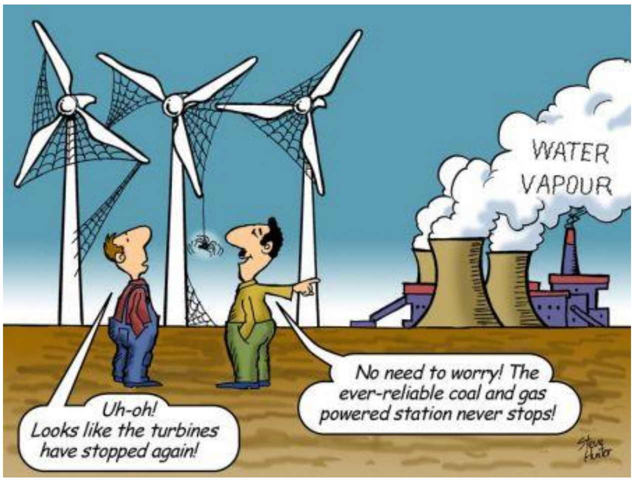
Be enough or adequate.

This means that to produce a significant amount of power, they need huge areas of land. A couple of examples will *suffice* to show the scale of this problem. If we used corn ethanol - a type of biofuel - to meet the world demand for energy, we would have to convert approximately half of the land surface of the entire planet into corn production. To keep up with only the world's *increase* in energy use, we would have to convert an area the size of Germany into wind farms *every year*. The practical consequences of this are obvious. For wind or biofuels to make an appreciable contribution to meeting our future energy needs, vast swathes of the country would have to be covered in turbines or corn plantations.

Figure 2. Land-use intensity



(C) Nuclear power is the only carbon-free energy source that can produce the vast amounts of energy we need without interruption. It can produce energy 24/7 for 365 days a year. It is reliable, predictable and _________- everything some of its competitors are not. Wind and solar power, for example, are dependent on the vagaries of the weather. If the wind does not blow or the sun does not shine, there is no energy produced. Another feature that must be factored in is the capacity factor (see fig 3) for wind and solar. Wind turbines and solar panels rarely operate at 100% efficiency. On many days the wind velocity is insufficient for the wind turbine to meet its optimum speed. Also, rather perversely, on days with high-velocity winds, the turbines have to be 'switched off' to prevent mechanical damage. As regards solar panels, cloud cover and a build-up of dirt on the panels can considerably reduce efficiency. This means that over a year a wind turbine or solar panel may only produce about 30% of its maximum theoretical output.



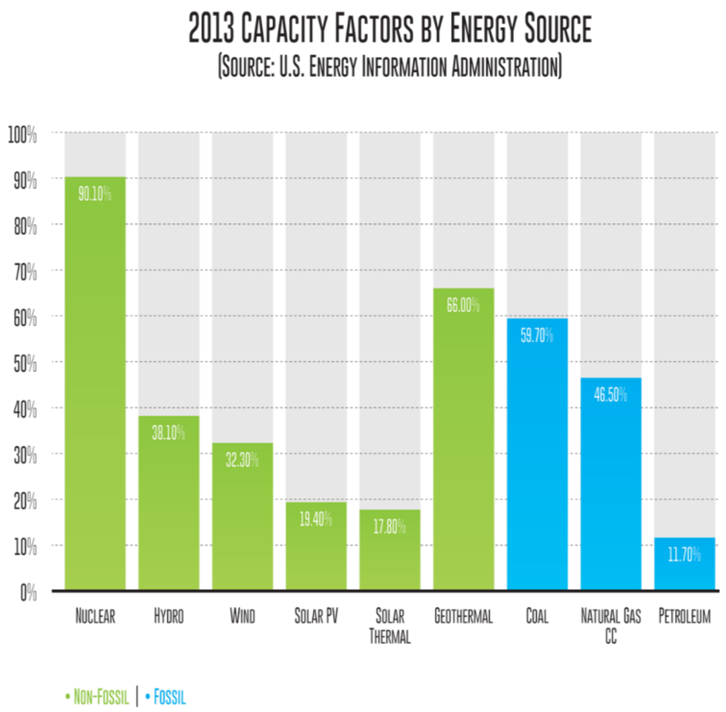
Capacity Factor

(D) The capacity factor is also linked to the previously discussed ideas of power density and intermittency. It is a major consideration when discussing the efficiency of energy sources in electricity generation. It is the ratio between the actual electricity production and maximum possible electricity output of a power plant over a period of time.

(E) Renewable power generation has a significantly lower capacity factor than baseload power plants due to the variability of the wind and sun. The baseload power plants - which use fuel sources such as nuclear, coal, natural gas or hydro - can operate continuously, unlike variable resources such as the wind and solar facilities. Often, when discussing power output of wind and solar, the figure given is maximum power output under optimal weather conditions. Unfortunately, due to the vagaries of the weather (cloud cover, variable wind speed, etc.), this maximum output is rarely achieved.

This is a very convincing reason of way we should not make solar and wind power our main energy source.

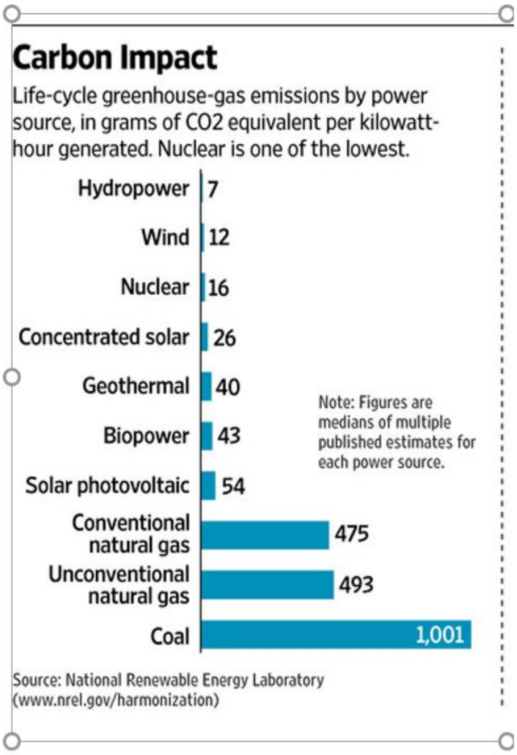
Figure 3.



CO2 emissions

(F) As we can see from Fig. 2, the burning of traditional fossil fuels for electricity generation produces a massive amount of CO2. In comparison, the CO2 produced from renewable energy sources is _____________. While this is obviously beneficial, it cannot be the only factor taken into account when choosing an energy source. If, for example, the comparison is with regard to electricity generated, the reverse is true. Output from renewables comes a distant second to that of fossil fuels. Again, nuclear power is the only source of energy that can reconcile massive energy output and low CO2 emissions.

Figure 4.



Nuclear Waste

(G) Aside from accidents, the public's negative perceptions of nuclear energy are with respect to the waste produced. The waste produced by a nuclear generating facility is highly radioactive. It has a relatively long life cycle and remains radioactive for thousands of years, which means it is a problem passed on to future generations. It requires special storage facilities, which can greatly add to its cost.

(H) However, it is important to mention that unlike fossil fuels, nuclear power generation fully regulates all waste. Nuclear power produces a relatively small quantity of highly toxic waste, which then must be carefully managed. However, the burning of fossil fuels produces huge amounts of waste in the form of greenhouse gases and air pollutants, which are then released directly into the environment.

Figure 5.

Why do you think the author includes this picture?

To show how long can the waste last and separate frome generation to another.

That means we cannot rely on it 100% without considering other sources.

(I) It is evident from the discussion above that nuclear power must be a significant part of our future energy mix. It is the only energy source that can meet the ever-increasing energy demand, without producing the huge quantities of greenhouse gases that are causing climate change. While renewables such as wind and solar have a role to play, it is a supporting role. Due to some of the weaknesses set out above, it would be a big mistake to think otherwise.

These two texts have an extremely useful writing structure, and it will probably help to develop our writing if we use it.

Paragraph A: Based on the text, complete the following sentences.

1. Although renewables have received significant investment it only produces 10% of the energy in US which is not enough for country

2. If we factor in population increase, we will realize that we shuold search for efficient energy source.

3. Nuclear power is the only realistic option because it does not require a lot of land and it does not provide emissions.

Complete the table using Figure 1.

Renewable Energy	__10__ %	wind	19%
		solar	6%
		biomass	49%
		geothermal	2%
		wood	21%
		hydroelectric	25%

4. Based on what you have read in Paragraph (B) (lines 13-24), would solar power be a realistic option in Saudi Arabia? Explain your answer.

Saudi Arabia has a large area of desert, and we can benefit from it if we use the desert to put the solar panels and as the sun is often shinning it will keep producing energy. Solar power can be especially useful for a county such as Saudi which has a hot wither, but we cannot rely just on it.

5. Given the poor power density of wind and solar we cannot rely on them to generate the energy.

Paragraph C: Based on the text, complete the following sentences.

Unlike nuclear power, wind and solar depend on the wither condion and the available of sun and wind.

Wind and solar seldom operate at 100% efficiency because the wither cannot be ideal all the time.

Given the intermittent nature of the wind and sun, they cannot work on their best because the wither condition is changing.

Unlike renewable energy sources, nuclear power does not depend on the wither, and it can opearate 100%

When discussing the efficiency of energy sources, nuclear energy is the best choice as it is available almost all the time.

Paragraph E: Complete the following sentences.

11. Unlike renewable energy sources, base-load power plants have a higher capacity factor as it does not depend on the wither change.

Paragraph F: Complete the following sentence.

12. When discussing the efficiency of energy sources, nuclear energy is the best choice as it is available almost all the time.

13. When choosing an energy source, there are two factors we need to take into account: the amount of CO2 it produced and the output energy.

Nuclear waste

1. A. The public's opinion of nuclear power is influenced by the waste it produces.
   B. The disposal of nuclear waste is a short-term problem.

a. Statement A is True and Statement B is False.
b. Statement B is True and Statement A is False.
c. Both statements are True.
d. Both statements are False.

2. A. Nuclear waste is very strictly controlled.
   B. Fossil fuel waste is very strictly controlled.

a. Statement A is True and Statement B is False.
b. Statement B is True and Statement A is False.
c. Both statements are True.
d. Both statements are False.

3. A. The author believes only nuclear can provide the power we need with no CO2 emissions.
   B. The author sees no need to continue with renewable energy sources.

a. Statement A is True and Statement B is False.
b. Statement B is True and Statement A is False.
c. Both statements are True.
d. Both statements are False.

Given what you have read **in this text and the previous text** and what you know about KSA, do you think nuclear power is the way forward for KSA? (Full paragraph. Dig Deep)

Despite the large amount of energy KSA gain from petroleum, it must search for another source of energy that cannot finish and can provide an adequate energy for the country. Nuclear energy can be the best type as it does not require a lot of land area and it does not provide emission. Moreover, using nuclear energy can improve the economy and develop the country in the field of energy and power. Taking the waste into account, Saudi can manage this issue with building storage facilities to control the waste. If Saudi take nuclear energy into account, the future for the country would be brighter.

1. Which of the following statements is TRUE?
a. The cause of climate change remains unclear.
b. Most scientists accept that humans have altered the climate.
c. Climate change is not an accepted scientific theory.
d. The only cause of climate change is burning fossil fuels.

2. In line 3 '*impetus*' means…
a. decision
b. complication
c. motivation
d. obstruction

3. What is the purpose of paragraph (A)?
a. Illustrate the weaknesses of renewables
b. set out the case for nuclear power
c. provide statistics about energy use
d. highlight the advantages of nuclear power

4. The author's position is …
a. for nuclear power.
b. against renewables.
c. against nuclear power.
d. not stated.

5. In 2015, what proportion of the energy used in the US comes from fossil fuels?

a. approximately 80%

b. just over half

c. almost ¾.

d. less than half

6. What % of the USA's renewable energy comes from wind?

a. 10%

b. 6%

c. 21%

d. 19%

7. Which renewable energy source contributes most to the USA's energy needs?

a. hydroelectric

b. biofuels

c. geothermal

d. wood

8. *This* in line 21 refers to …

a. land requirements of renewables.

b. producing significant amounts of power.

c. the low power density of renewables.

d. limitations to energy output.

9. In line 22 '*suffice* 'means to be _____

a. inadequate

b. enough

c. unlikely

d. insufficient

10. Why is corn ethanol mentioned in the text?

a. because it is the most inefficient renewable.

b. to show corn needs huge areas of land.

c. as an example of a renewable energy.

d. as an illustration of the poor power density of renewables.

11. According to the text, which fossil fuel has the lowest power density?

a. Oil

b. Gasoline

c. Coal

d. Methane hydrates

12. The purpose of paragraph (B) is to…

a. explain why nuclear is the more efficient.

b. show renewables are not a practical option.

c. compare power densities of energy sources.

d. set out the inefficiencies of corn-ethanol.

13. Which renewable energy is the most efficient with regard to land use?

a. biomass
b. geothermal
c. solar thermal
d. hydro

14. What is the best sub-heading for paragraph (C)?
a. Blowing in the wind
b. Inefficiencies of wind
c. Intermittency
d. Variation in weather

15. Which of the following would best fit in the space in line 28?
a. consistent
b. expensive
c. efficient
d. variable

16. From paragraph (C) we can infer we should…
a. build more wind and solar to improve output.
b. ignore the maximum output claims of wind and solar manufacturers.
c. locate wind and solar in windier, sunnier places.
d. forget renewables and stay with fossil fuels.

17. Which of the following statements is TRUE?
a. Baseload power can only be provided by fossil fuels.
b. Hydropower can not be used for baseload power.
c. Baseload refers to maximum output in all weather conditions.
d. Baseload power is not affected by variations in the weather.

18. When discussing the maximum output of wind and solar…
a. the figure given is average output.
b. the figure takes into account weather variations.
c. the capacity factor is not usually considered.
d. the output is only an estimate.

19. Which fossil fuel has the best capacity factor?
a. petroleum
b. coal
c. natural gas
d. nuclear

20. Which non-fossil fuel has the best capacity factor?
a. geothermal
b. hydro
c. natural gas
d. nuclear

21. Which of the following best fits into the space in line 65?
a. insignificant
b. considerable

c. surprising
d. unconnected

22. The energy source with the lowest carbon impact is…
a. coal
b. conventional natural gas
c. petroleum
d. hydropower

23. With regard to nuclear power, what does the public worry about most?
a. waste
b. accidents
c. waste disposal
d. terrorism

24. The problem highlighted in paragraph (G) is…
a. the dangerous nuclear waste produced.
b. the expense involved in building storage facilities.
c. the length of time waste must be stored.
d. the difficulty finding suitable storage sites.

25. Which of the following statements is TRUE?
a. Nuclear produces more waste than fossil fuels.
b. Waste from fossil fuels is carefully controlled.
c. Fossil fuel waste is more dangerous than nuclear waste.
d. All nuclear waste is carefully controlled.

26. What does the author NOT do in the final paragraph?
a. restates his thesis
b. rejects the use of renewables
c. look to the future
d. state his position